

GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 7

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Work out $\frac{1}{4}$ of 60.

(Total for Question 1 is 1 marks)

2.

A jumper costs £45 and is reduced by 30% in a sale. Work out the sale price.

(Total for Question 2 is 2 marks)

3.

- (a) Round 7.39 to the nearest whole number. (1)
- (b) Round 234 to the nearest 10. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 18 : 42 in its simplest form.

(Total for Question 4 is 2 marks)

5.

Two ordinary fair six-sided dice are rolled.

- (a) How many possible outcomes are there in total? (1)
- (b) Work out the probability that the two dice show the same number. (2)

(Total for Question 5 is 3 marks)

6.

Expand $5(3x - 2)$.

(Total for Question 6 is 2 marks)

7.

A rectangular field is 50 m long and 30 m wide. Sam walks from one corner of the field to the opposite corner in a straight line.

[Diagram: rectangle 50 m by 30 m with a diagonal drawn corner-to-corner.]

Work out the length of the diagonal walk. Give your answer to 1 decimal place.

(Total for Question 7 is 3 marks)

8.

The table shows the number of children per household for 20 households in a street.

Children	0	1	2	3	4
Freq.	4	6	7	2	1

Work out the mean number of children per household. Give your answer to 2 decimal places.

(Total for Question 8 is 3 marks)

9.

Town B is due east of Town A. Town C is on a bearing of 070 deg from Town A.

[Diagram: point A at origin; B is due east; C is in the NE quadrant such that the angle from North-line at A to AC measured clockwise is 70 deg.]

(a) Write down the bearing of B from A. (1)

(b) Work out the angle BAC. (2)

(Total for Question 9 is 3 marks)

10.

Solve $7x - 4 = 24$.

(Total for Question 10 is 3 marks)

11.

A cuboid has dimensions 12 cm by 5 cm by 4 cm.

(a) Work out the volume of the cuboid. (2)

(b) Work out the length of one of the longest edges. (You may state the answer.) (1)

(Total for Question 11 is 3 marks)

12.

Layla puts £5000 in a savings account paying 3.2% compound interest per year.

Work out the value of her investment after 2 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

A bag contains 8 sweets: 3 toffees and 5 mints. A sweet is taken at random, the colour noted, and then replaced. A second sweet is then taken.

- (a) Write down the probability of taking a toffee on a single pick. (1)
- (b) Work out the probability that both sweets taken are toffees. (3)

(Total for Question 13 is 4 marks)

14.

An isosceles triangle has a base of 16 cm and two equal sides of length 17 cm.

[Diagram: isosceles triangle with base 16 cm and two equal sides of 17 cm; perpendicular height from apex to base labelled h .]

- (a) Show that the perpendicular height h of the triangle is 15 cm. (3)
- (b) Hence work out the area of the triangle. (2)

(Total for Question 14 is 5 marks)

15.

A closed cuboid has dimensions 8 cm by 5 cm by 6 cm.

- (a) Work out the volume of the cuboid. (2)
- (b) Work out the total surface area of the cuboid. (3)

(Total for Question 15 is 5 marks)

16.

The n th term of a sequence is given by $T_n = 3n + 5$.

- (a) Write down the first three terms of the sequence. (2)
- (b) Work out T_{20} . (1)
- (c) Is 62 a term of the sequence? Show your working. (1)

(Total for Question 16 is 4 marks)

17.

The table shows the height (cm) and weight (kg) of 8 students.

Height	150	155	158	162	166	170	174	180
Weight	48	52	55	58	62	65	70	75

- (a) On the grid provided, plot these data as a scatter graph. (2)
- (b) Describe the correlation. (1)
- (c) Draw a line of best fit. (1)
- (d) Estimate the weight of a student who is 165 cm tall. (1)

(Total for Question 17 is 5 marks)

18.

A triangular prism has a cross-section that is a right-angled triangle with shorter sides of 6 cm and 8 cm. The length of the prism is 20 cm.

[Diagram: right-angled triangular prism: cross-section right-angled triangle with shorter sides 6 cm and 8 cm; length of prism 20 cm.]

- (a) Work out the volume of the prism. (3)

The prism is made of a metal of density 2.7 g/cm^3 .

- (b) Work out the mass of the prism. Give your answer in kilograms. (2)

(Total for Question 18 is 5 marks)

19.

A map of a country park uses a scale of 1 cm to 200 m.

- (a) Write the scale of the map in the form 1 : n. (2)
- (b) The distance between the visitor centre and the lake is 6.5 cm on the map. Work out the real distance in metres. (2)
- (c) The real distance between the lake and the play area is 1.4 km. Work out the distance on the map in centimetres. (1)

(Total for Question 19 is 5 marks)

20.

Asha and Ben share some money in the ratio 4 : 5. Ben receives £25 more than Asha. Let the amount Asha receives be £x.

- (a) Show that $x = 100$. (3)
- (b) Work out the total amount of money they share. (2)

(Total for Question 20 is 5 marks)

21.

Priya invests £8000 in a savings plan.

For the first 2 years it pays compound interest at 4% per year.

In year 3 the rate falls to 2% per year.

- (a) Work out the value of the investment at the end of year 2. (3)
- (b) Work out the value at the end of year 3. Give your answer to the nearest penny. (3)

(Total for Question 21 is 6 marks)

22.

60 students in a sixth-form chose one of three sports. The two-way table shows some information.

	Football	Hockey	Tennis	Total
Boys	16	7	—	30
Girls	—	9	13	—

Total _____ 60

- (a) Complete the two-way table. (3)
- (b) A student is chosen at random. Work out the probability that the student is a girl who chose Football. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 7 | 80 marks

Question 1

- 15 - B1

Total: 1 marks

Question 2

- Multiplier 0.7 or 30% of 45 = 13.50 seen - M1
- £31.50 - A1

Total: 2 marks

Question 3

- (a) 7 - B1
- (b) 230 - B1

Total: 2 marks

Question 4

- Both parts divided by HCF (e.g. 6:14 then 3:7) - M1
- 3 : 7 - A1

Total: 2 marks

Question 5

- (a) 36 - B1
- (b) Six "doubles" (1,1)(2,2)...(6,6) identified - M1
- (b) $6/36 = 1/6$ - A1

Total: 3 marks

Question 6

- First multiplication correct: $5 \times 3x = 15x$ or $5 \times (-2) = -10$ - M1
- $15x - 10$ - A1

Total: 2 marks

Question 7

- $d^2 = 50^2 + 30^2$ - M1
- $2500 + 900 = 3400$ - M1
- 58.3 (m) to 1 dp - A1

Total: 3 marks

Question 8

- $\text{Sum}(fx) = 0 + 6 + 14 + 6 + 4 = 30$ - M1
- $30 / 20$ - M1
- 1.50 - A1

Total: 3 marks

Question 9

- (a) 090 (deg) - B1
- (b) 90 - 70 attempted - M1
- (b) 20 (deg) - A1

Total: 3 marks

Question 10

- $7x = 28$ - M1
- $x = 4$ written - M1
- $x = 4$ with working clear - A1

Total: 3 marks

Question 11

- (a) $12 \times 5 \times 4$ attempted - M1
- (a) $240 \text{ (cm}^3\text{)}$ - A1
- (b) 12 (cm) - B1

Total: 3 marks

Question 12

- Multiplier 1.032 seen - M1
- $5000 \times 1.032 = 5160$ (year 1) - M1
- 5160×1.032 - M1
- £5325.12 - A1

Total: 4 marks

Question 13

- (a) $\frac{3}{8}$ - B1
- (b) Multiplication of independent events - M1
- (b) $\frac{3}{8} \times \frac{3}{8}$ - M1
- (b) $\frac{9}{64}$ - A1

Total: 4 marks

Question 14

- (a) Half the base = 8 cm identified - M1
- (a) $h^2 = 17^2 - 8^2 = 289 - 64 = 225$ - M1
- (a) $h = \sqrt{225} = 15 \text{ (cm)}$ shown - C1
- (b) Area = $0.5 \times 16 \times 15$ - M1
- (b) $120 \text{ (cm}^2\text{)}$ - A1

Total: 5 marks

Question 15

- (a) $8 \times 5 \times 6$ attempted - M1
- (a) $240 \text{ (cm}^3\text{)}$ - A1
- (b) $2(8 \times 5) + 2(8 \times 6) + 2(5 \times 6)$ - M1
- (b) $80 + 96 + 60$ - M1
- (b) $236 \text{ (cm}^2\text{)}$ - A1

Total: 5 marks

Question 16

- (a) Substitute $n = 1, 2, 3$ - M1
- (a) 8, 11, 14 - A1
- (b) $3 \times 20 + 5 = 65$ - B1
- (c) $3n + 5 = 62 \rightarrow n = 19$ (an integer) so YES 62 is a term - B1

Total: 4 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 if 6+)
- (b) Positive correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Weight at 165 cm (accept 58 - 64 kg) - B1

Total: 5 marks

Question 18

- (a) $0.5 \times 6 \times 8 = 24$ (area of triangle) - M1
- (a) 24×20 - M1
- (a) $480 \text{ (cm}^3\text{)}$ - A1
- (b) $2.7 \times 480 (= 1296 \text{ g})$ - M1
- (b) 1.296 (kg) - A1

Total: 5 marks

Question 19

- (a) $200 \text{ m} = 20\,000 \text{ cm}$ - M1
- (a) $1 : 20\,000$ - A1
- (b) 6.5×200 - M1
- (b) 1300 (m) - A1
- (c) $1.4 \text{ km} = 1400 \text{ m}; / 200 = 7 \text{ cm}$ - B1

Total: 5 marks

Question 20

- (a) Asha = x , so one part = $x/4$ - M1
- (a) Ben = 5 parts = $5x/4 = x + 25$ - M1
- (a) $x/4 = 25 \rightarrow x = 100$ shown - C1
- (b) Asha £100, Ben = $5/4 \times 100 = £125$ - M1
- (b) Total = $100 + 125 = £225$ - A1

Total: 5 marks

Question 21

- (a) Multiplier 1.04 seen - M1
- (a) $8000 \times 1.04 = 8320$ (year 1) seen - M1
- (a) $8320 \times 1.04 = £8652.80$ - A1
- (b) Multiplier 1.02 used for year 3 - M1
- (b) 8652.80×1.02 - M1

- (b) £8825.86 (accept £8825.856 to 8825.86) - A1

Total: 6 marks

Question 22

- (a) Boys Tennis = $30 - 16 - 7 = 7$ - M1
- (a) Girls total = $60 - 30 = 30$; Girls Football = $30 - 9 - 13 = 8$ - M1
- (a) Column totals: Football 24, Hockey 16, Tennis 20 - A1
- (b) Girls Football = 8 out of 60 - M1
- (b) $8/60 = 2/15$ - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 7 | Detailed explanations

Question 1

$\frac{1}{4}$ means divide by 4.

$$60 \div 4 = 15.$$

Question 2

30% off means the sale price is 70% of the original.

$$0.7 \times 45 = 31.50.$$

Answer: £31.50.

Question 3

(a) 7.39: the next digit (tenths) is 3 (< 5), so round down. Answer: 7.

(b) 234: the units digit is 4 (< 5), so round down. Answer: 230.

Question 4

HCF of 18 and 42 is 6. Divide both parts: $18/6 = 3$, $42/6 = 7$.

Answer: 3 : 7.

Question 5

(a) Each die has 6 outcomes, so two dice give $6 \times 6 = 36$ ordered outcomes.

(b) "Same number" means doubles: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6). That is 6 outcomes.

$$P(\text{same number}) = 6 / 36 = 1/6.$$

Question 6

Multiply each term inside the bracket by 5.

$$5 \times 3x = 15x. \quad 5 \times (-2) = -10.$$

Answer: $15x - 10$.

Question 7

The diagonal is the hypotenuse of a right-angled triangle with legs 50 m and 30 m.

$$d^2 = 50^2 + 30^2 = 2500 + 900 = 3400.$$

$$d = \sqrt{3400} = 58.309..., \text{ i.e. } 58.3 \text{ m to 1 dp.}$$

Question 8

$$\text{Sum}(fx) = 0 \times 4 + 1 \times 6 + 2 \times 7 + 3 \times 2 + 4 \times 1 = 0 + 6 + 14 + 6 + 4 = 30.$$

$$\text{Sum}(f) = 20.$$

Mean = $30 / 20 = 1.50$.

Question 9

- (a) Due east is 090° deg.
(b) Angle BAC = bearing of B - bearing of C = $90 - 70 = 20^\circ$ deg.
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Question 10

$7x - 4 = 24$. Add 4: $7x = 28$. Divide by 7: $x = 4$.
Check: $7(4) - 4 = 28 - 4 = 24$.

Question 11

- (a) $V = 12 \times 5 \times 4 = 240 \text{ cm}^3$.
(b) The longest edge is the longest dimension - 12 cm.
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Question 12

Multiplier = 1.032.
Year 1: $5000 \times 1.032 = \text{£}5160.00$.
Year 2: $5160 \times 1.032 = \text{£}5325.12$.

Question 13

- (a) $P(\text{toffee}) = 3/8$.
(b) After replacement, the second pick is independent of the first.
 $P(\text{both toffees}) = (3/8) \times (3/8) = 9/64$.
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Question 14

- (a) Drop the perpendicular from the apex to the base. It bisects the base, giving a right-angled triangle with hypotenuse 17 and one shorter side 8.
 $h^2 = 17^2 - 8^2 = 289 - 64 = 225$.
 $h = \sqrt{225} = 15 \text{ cm}$. (Shown.)
(b) Area = $0.5 \times \text{base} \times \text{height} = 0.5 \times 16 \times 15 = 120 \text{ cm}^2$.
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Question 15

- (a) $V = 8 \times 5 \times 6 = 240 \text{ cm}^3$.
(b) Three pairs of identical rectangular faces.
 $2(8 \times 5) + 2(8 \times 6) + 2(5 \times 6) = 80 + 96 + 60 = 236 \text{ cm}^2$.
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Question 16

- (a) $T_1 = 3(1) + 5 = 8$. $T_2 = 3(2) + 5 = 11$. $T_3 = 3(3) + 5 = 14$.
(b) $T_{20} = 3(20) + 5 = 65$.
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(c) $3n + 5 = 62 \rightarrow 3n = 57 \rightarrow n = 19$. n is a whole number, so 62 is the 19th term. Yes.

Question 17

- (a) Plot 8 points.
 - (b) Positive correlation - taller students tend to be heavier.
 - (c) Best fit drawn.
 - (d) At 165 cm, line gives roughly 58 - 64 kg.
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Question 18

- (a) Cross-section is right-angled triangle: $\text{area} = 0.5 \times 6 \times 8 = 24 \text{ cm}^2$.
 $\text{Volume} = \text{area} \times \text{length} = 24 \times 20 = 480 \text{ cm}^3$.
 - (b) $\text{Mass} = \text{density} \times \text{volume} = 2.7 \times 480 = 1296 \text{ g}$.
Convert to kg: $1296 / 1000 = 1.296 \text{ kg}$.
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Question 19

- (a) 1 cm represents 200 m. Convert 200 m to cm: $200 \times 100 = 20\,000 \text{ cm}$. So the scale is 1 : 20 000.
 - (b) 6.5 cm on map = $6.5 \times 200 = 1300 \text{ m}$ in real life.
 - (c) 1.4 km = 1400 m. On map: $1400 / 200 = 7 \text{ cm}$.
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Question 20

- (a) Asha gets x (i.e. 4 parts), so one part = $x / 4$.
Ben gets 5 parts = $5 \times (x/4) = 5x/4$.
 $\text{Ben} - \text{Asha} = 5x/4 - x = (5x - 4x)/4 = x/4 = 25$.
So $x = 100$. (Shown.)
 - (b) Asha £100, Ben £125. Total = £225.
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Question 21

- (a) Years 1 and 2 multiplier = 1.04.
Year 1: $8000 \times 1.04 = \text{£}8320.00$.
Year 2: $8320 \times 1.04 = \text{£}8652.80$.
 - (b) Year 3 multiplier = 1.02.
Year 3: $8652.80 \times 1.02 = \text{£}8825.856$, i.e. £8825.86 to the nearest penny.
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Question 22

- (a) Boys row total 30, so Boys Tennis = $30 - 16 - 7 = 7$.
Girls row total = $60 - 30 = 30$, so Girls Football = $30 - 9 - 13 = 8$.
Column totals: Football $16+8=24$, Hockey $7+9=16$, Tennis $7+13=20$. Check: $24+16+20=60$.
 - (b) $P(\text{girl AND Football}) = 8/60 = 2/15$.
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